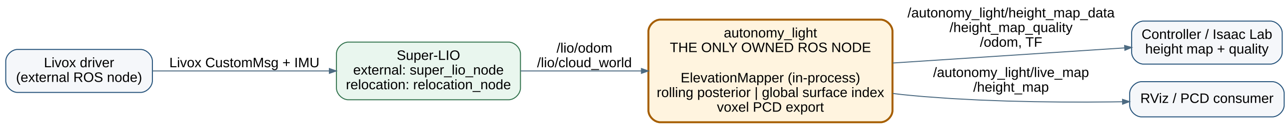
autonomy_light software architecture

Super-LIO integration and elevation mapping · 2026-08-02

> **Design decision.** SLAM and relocalization are Super-LIO responsibilities. This package owns one ROS node only: it converts Super-LIO output into a robust elevation map and a sparse PCD.



Node responsibilities

Node	Owner	Responsibility
Livox driver	external	Publishes one Livox CustomMsg and IMU stream.
Super-LIO	external	LiDAR–IMU odometry, registered global cloud, and saved-map relocation.
`autonomy_light`	this package	One node: base pose/TF, rolling or global elevation map, quality output, live PCD and PCD save.

Data and TF contract

Direction	Topic	Meaning
input	`/lio/odom`	Super-LIO LiDAR pose; converted to `base_link`.
input	`/lio/cloud_world`	Registered cloud in the current global frame.
output	`/autonomy_light/height_map_data`	Controller distance grid.
output	`/autonomy_light/height_map_quality`	Per-cell validity, variance, age.
output	`/autonomy_light/live_map`	Sparse voxel PCD for RViz and map saving.

TF is `odom|map → base_link → base_link_gravity` and static `base_link → lidar_link`. Normal operation uses `odom`; a saved map runs Super-LIO `relocation_node` and uses `map`. There is no `world` frame and no locally computed `map → odom` correction.

Elevation source selection

Configuration	Algorithm	Use
`height_map.source: rolling`	Bounded Bayesian ground/upper posterior with measurement variance, aging and outlier gate.	Default for control and sim-to-real.
`height_map.source: global`	Persistent, PCD-derived sparse surface index.	Repeatable terrain reference from a saved or accumulated map.

Both sources use the same output message. `valid=0` means the distance value is an unknown placeholder, not measured flat ground; Isaac Lab should consume the quality channels.

Removed ownership

- Local pose graph, Scan Context loop closure, and map reconstruction
- FPFH/GICP initial localization and runtime map correction
- Secondary LiDAR merger and custom ROS-domain bridge executors
- Separate raw/refined/ROI map pipelines and duplicated path subscription

These removals make the state path linear: Super-LIO localizes; one node maps elevation.